

Affording Generative AI Across Countries: Uniform Prices, Unequal Burdens

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Abstract

Consumer generative AI is sold on a subscription menu that is nearly identical across countries, while national incomes differ by two orders of magnitude. This paper measures how far local prices adjust to that difference. I audit the tax-inclusive local prices listed for nine plans of ChatGPT, Claude and Gemini in 170 national Apple App Store storefronts on a single day, and compute the annual cost of each plan as a share of GNI per capita for 201 economies. Standard \$20 plans are priced almost uniformly: the income slope of the local dollar price is 0.03–0.04, against 0.23 if prices tracked local price levels. Only OpenAI’s entry tier is partly localised (slope 0.14), and every low-income storefront prices in U.S. dollars. A \$20 plan therefore costs 0.65% of per-capita income a year in the median high-income economy and 29% in the median low-income economy, and 60% of the working-age population of the covered economies lives where it exceeds the 2% benchmark used for broadband. Two public usage indicators differ in their income gradients, 0.37 for any generative-AI use and 0.71 for per-capita Claude activity, a contrast consistent with, though not a test of, a model in which free tiers detach initial adoption from the paid price. Larger productivity gains for less-skilled users, moreover, do not imply convergence across countries unless exposure and adoption keep pace. The audit is a one-day snapshot of listed prices; the paper sets out what the evaluation of pricing and access policies would require.

Keywords: generative AI; affordability; digital divide; international price discrimination; technology diffusion; development.

JEL codes: O33, O15, L11, L86, D63, F63.

1 Introduction

Field and experimental evidence suggests that generative AI raises productivity, with the largest gains often accruing to less-experienced workers (Brynjolfsson et al., 2025; Noy and Zhang, 2023; Dell’Acqua et al., 2023; Peng et al., 2023). Those findings concern people who already had access. Whether the same opportunity exists across countries depends first on whether people can use the tools, and frontier capability is sold to individuals as a subscription on a menu that is standardised worldwide: a free tier, a standard tier at \$20 a month and high-usage tiers at about \$100 and \$200 (Table 1). The same dollar price weighs very differently on different incomes. Twenty dollars a month is 0.3% of per-capita income in the United States and more than 40% in Mozambique.

This paper asks how far local subscription prices adjust to differences in national income, and what a uniform price implies for the affordability of generative AI across countries. Its contribution is a dated, multi-provider audit of local prices: the tax-inclusive prices listed for nine plans of ChatGPT, Claude and Gemini in every Apple App Store storefront (218 codes, 170 with local price lists), collected on 27 September 2026 through a single payment channel, together with the annual cost of each plan as a share of GNI per capita for 201 economies. Earlier work compared ChatGPT Plus’s \$20 fee with national incomes (Sathish et al., 2024); this audit adds local rather than U.S. prices, several providers and tiers, and a record of retrieval outcomes, currencies and plan matching. A matched comparison of ChatGPT Go and Plus within each storefront checks that the entry-tier result does not depend on currency conversion.

Four findings follow. *Standard plans are priced almost uniformly.* Across storefronts the income slope of the local dollar price is 0.039 for ChatGPT Plus, 0.026 for Claude Pro and 0.037 for Google AI Pro, against 0.23 if prices tracked local price levels and 1 under proportional pricing. *Only OpenAI’s entry tier is partly localised.* ChatGPT Go’s slope is 0.143, about 60% of the price-level slope and one-seventh of proportional pricing; Google’s entry tier is listed at \$4.99 in every non-high-income group. Every storefront in a low-income economy prices in U.S. dollars, and 36% of low-income economies have no local storefront at all. *Burdens are therefore steeply income-graded.* A \$20 plan costs 0.65% of per-capita income a year in the median high-income economy, 3.1% in upper-middle-, 8.7% in lower-middle- and 29% in low-income economies. It meets the 2% benchmark that the ITU and the UN Broadband Commission apply to broadband only where per-capita income is at least \$12,000, and 60% of the working-age population of the covered economies lives where it does not. For the richest fifth of households in the median low-income economy, one plan would still cost 13% of average income. *Public usage indicators differ in their income gradients.* Microsoft’s estimate of the share of adults using any generative-AI product has an income elasticity of 0.37, about half of which coincides with the gradient in internet use; Anthropic’s index of per-capita Claude use has an elasticity of 0.71 on the same 96 economies, of which only one is low-income.

A simple framework organises these facts. In a three-option model of non-use, free use and paid use, a free tier detaches the decision to use AI at all from the paid price whenever the upgrade threshold lies above the free-adoption threshold; the paid margin then inherits the full income gradient unless prices are localised (Appendix A). The steeper gradient of Claude activity is consistent with this mechanism, but cross-country data cannot identify it: vendor choice, conversations per user and measurement processes can produce the same ordering, and the index pools free and paid accounts. A second result concerns convergence. Because aggregate gains depend on exposure and adoption as well as on task-level gains,

larger conditional gains for less-skilled users are neither necessary nor sufficient for AI to narrow gaps between countries (Proposition 1). On rough proxies for exposure and adoption, low-income adoption would have to reach about 100% for equal task-level gains to narrow the gap, and 50–67% even if gains there were 1.5–2 times larger, against about 9% today.

The paper relates to several literatures. Studies of who adopts generative AI within countries find adoption skewed toward the young, the educated and higher earners (Bick et al., 2024; Humlum and Vestergaard, 2025); cross-country studies based on web traffic, telemetry or conversation data document large gaps by income (Liu and Wang, 2024; Liu et al., 2025; Misra et al., 2025; Chatterji et al., 2025; Appel et al., 2025; Daepp and Slaughter, 2026; Iscenko et al., 2026); and direct surveys are beginning to distinguish free from paid use (Kanzamanova and Myeong, 2026). Earlier work on computers, the internet and mobile telephony identified income, human capital and prices as the main correlates of diffusion (Caselli and Coleman, 2001; Chinn and Fairlie, 2007; Aker and Mbiti, 2010; Björkegren, 2019), and the diffusion literature has long distinguished the timing of adoption from the intensity of use (Comin and Hobijn, 2004, 2010): Comin and Mestieri (2018) show that for earlier technologies adoption lags converged across countries while usage intensity diverged. Studies of occupational exposure to AI (Felten et al., 2021; Eloundou et al., 2024; Cazzaniga et al., 2024; Gmyrek et al., 2025) supply the exposure figures used below, and development-oriented assessments (Korinek and Stiglitz, 2021; United Nations Development Programme, 2025; UNCTAD, 2025; World Bank, 2026c) frame the policy stakes. The affordability convention for telecommunications (International Telecommunication Union, 2025a) supplies the benchmark extended here, and the literature on international pricing and product versioning (Goldberg and Knetter, 1997; Cavallo et al., 2014; Mussa and Rosen, 1978; Goldfarb and Tucker, 2019) frames the audit, although taxes, platform commissions and product differences mean that the estimated slopes are not firms’ price-discrimination parameters.

The analysis is descriptive. Because a uniform dollar price makes the burden a monotone transformation of income, a cross-country association between burdens and use cannot identify a price effect; the audit also postdates the usage windows and observes one channel on one day. The staggered rollout of localised entry tiers, and the tariff change that OpenAI announced two days after the audit, now make evaluation designs feasible, and Section 6 sets out what they require.

Section 2 describes the price menu and the subsequent tariff change. Section 3 defines the burden measure and presents the framework. Section 4 documents the data, Section 5 reports the audit, the burdens and the usage comparison, Section 6 discusses interpretation, evaluation designs, policy and limitations, and Section 7 concludes. The formal model, data construction, a connectivity decomposition, additional tables, a \$500 reference-fee scenario and a measurement protocol are in the appendices.

2 The price menu

Table 1 summarises the consumer plans of the three leading assistants as documented on vendors’ official pages on 27 September 2026, together with the U.S. App Store price of the same plans. All price regressions use the App Store snapshot of that day; the usage windows analysed later end in May or June 2026, so the audited prices postdate the usage data. Three features of the menu stand out.

Convergence. The three menus have converged on nearly the same rungs: a free tier, an entry tier at \$5–8, a \$20 standard tier, and tiers at about \$100 and \$200. Using several assistants multiplies the cost: a professional who keeps the \$20 tier of all three pays about \$60 a month.

Table 1: The consumer price ladder of three leading AI assistants, September 2026

Tier	OpenAI (ChatGPT)	Anthropic (Claude)	Google (Gemini)
Free	Free	Free	Free
Entry	Go: \$8 [\$8.00]; launched in India Aug. 2025, 170 further countries, U.S. Jan. 2026; ads tested	—	AI Plus: [\$4.99]
Standard	Plus: \$20 [\$19.99]	Pro: \$20 monthly, \$17 annual [\$20.00]	AI Pro: [\$19.99]
High usage	Pro \$100: 5× Plus usage [\$100]	Max: “from \$100”, 5× or 20× Pro usage [\$124.99; \$249.99]	AI Ultra \$100: 5× Pro usage
Highest usage	Pro \$200: 20× Plus usage [\$200]; new sign-ups paused since 10 Sep. 2026	(Max 20×, see above)	AI Ultra \$200 (was \$250): 20× Pro usage [\$199.99]

Notes: Official U.S. web prices per month as documented on 27 September 2026, with U.S. App Store prices in brackets (same day, before sales tax). Web prices and allowance multiples supply context; the cross-country regressions use only the App Store records. OpenAI: “Both Pro tiers include the same core capabilities. The main difference is usage allowance” (OpenAI, 2026a); Plus price and separate API billing from OpenAI (2026e); Go from OpenAI (2026d). Anthropic: Anthropic (2026b) (prices exclude tax). Google: new \$100 Ultra tier and \$200 Ultra (down from \$250) announced 19 May 2026 (Google, 2026). Google’s plan page displays region-dependent prices, so bracketed App Store prices are shown for AI Plus and AI Pro. The OpenAI terms changed on 29 September 2026 (Section 2.1).

Quantity, not intelligence. The upper rungs are differentiated primarily by usage allowances (“5×” and “20×” the standard tier) and, secondarily, by access to the most capable model variants and features. With near-zero marginal cost of replication but non-trivial inference costs, digital goods lend themselves to versioning (Goldfarb and Tucker, 2019), and the menu is a textbook instance of second-degree price discrimination by quantity and quality (Mussa and Rosen, 1978). The free and entry tiers are degraded versions of the standard tier: ChatGPT Go, for instance, offers an “Instant” model with lower caps, while Plus offers the “Thinking” model with higher limits (OpenAI, 2026d). A higher price therefore buys more frontier capability per month, although neither the plan name nor the price is a comparable measure of useful capability across vendors, languages or tasks.

Localisation happens at the bottom of the ladder. The entry tiers are the vendors’ instrument for reaching lower-income markets: ChatGPT Go launched in India in August 2025 and was rolled out to 170 further countries before its U.S. launch at \$8 in January 2026 (OpenAI, 2026d). Section 5.2 shows that it is the only rung whose local price varies materially with income. Promotional give-aways, such as bundles of premium plans with mobile subscriptions in India, further lower effective prices in some markets, but they are temporary and country-specific and are not captured by a listing audit.

2.1 The tariff change of 29 September 2026

Two days after the audit, OpenAI announced a Pro 500 tier at \$500 a month, sold initially on the web and including its fastest service mode, and reopened Pro 200 subscriptions with a lower included allowance for new subscribers, while existing subscribers keep their allowance until 29 October 2026 (OpenAI, 2026c,b). The episode illustrates two measurement points. An unchanged fee can correspond to different entitlements by account cohort and date, and a new tier can add eligibility rights rather than replace a price. A tariff is therefore a bundle $(F, \mathbf{A}, \mathcal{E}, \mathbf{r}, \mathbf{w})$ of fee, included allowances, eligible models and service modes, overage terms and reset rules, of which the burden defined below uses the fee F alone.

Quality-adjusted prices in the model market fall rapidly (André et al., 2025); they are a different object from the recurring cash commitment of a subscription, and neither identifies the cash cost of completing a given task. The change postdates both the price audit and the usage windows, so it is treated separately: Appendix E reports a \$500 reference-fee scenario, and Appendix F sets out a protocol for recording the elements of a tariff that the fee alone omits.

3 Framework

3.1 The affordability burden

Definition 1 (AI affordability burden). For a plan with monthly price p_c in country c (current U.S. dollars) and per-capita income y_c (current U.S. dollars),

$$AAB_c(p_c) = \frac{12p_c}{y_c} \times 100. \quad (1)$$

The burden is a normalised price: it states how large the price is relative to average income. It is not a budget share and does not measure what anyone spends. The denominator is Atlas-method GNI per capita, which smooths exchange rates over three years. The measure follows the convention of the ITU’s broadband affordability statistics, which express the monthly price of an entry-level basket as a share of monthly GNI per capita and assess it against the 2% benchmark of the UN Broadband Commission (International Telecommunication Union, 2025a). I use that benchmark as a borrowed reference and report sensitivity to other cutoffs. For a cutoff t in percentage points the threshold income is $y_c \geq 1200p/t$; at 2% this is \$12,000 for a \$20 plan and \$4,800 for an \$8 plan.

Because average income conceals distribution, I also compute burdens at quintile means, approximating the mean income of quintile q by $y_{cq} = y_c s_{cq}/0.2$, where s_{cq} is the quintile’s survey-based income or consumption share. These are synthetic resource levels rather than measured disposable incomes, since survey welfare concepts, survey dates and national accounts differ. GNI also includes income that does not accrue to households, so burdens relative to household final consumption per capita are reported as well.

Remark 1 (What a PPP denominator measures). For a service priced in dollars, the burden relative to income at market exchange rates is the relevant cash-cost quantity. Dividing the dollar price by income at purchasing-power parity instead yields $12p/y_c^{PPP} = 12(p \cdot PLR_c)/y_c$, where $PLR_c = y_c/y_c^{PPP}$, the ratio of Atlas to PPP income, is a proxy for the price level relative to the United States.¹ The same algebra equals the Atlas-income burden under a constructed price $p_c = p \cdot PLR_c$, that is, the burden that would obtain if the vendor indexed its price to local price levels. Because price levels rise with income (Balassa, 1964; Samuelson, 1964), and PPP conversion factors are designed for broad consumption baskets rather than internationally priced digital services (Deaton and Heston, 2010), the PPP-based ratio is best read as this counterfactual pricing scenario rather than as a robustness check on the same quantity. Section 5.2 compares actual pricing with it.

¹Because the Atlas method averages exchange rates over three years, the ratio differs from the same-year GDP price-level ratio by a median of 3.1% (90th percentile 8.3%) across 197 economies. Its income elasticity is 0.228, against 0.224 for the GDP price-level ratio.

3.2 A free tier and two margins of access

A free tier permits initial use while a paid tier offers more capability. In the three-option model of Appendix A, connected individuals with heterogeneous valuations choose non-use, free use or paid use. When the threshold for upgrading lies above the threshold for using the free tier, the decision to use AI at all is governed by non-monetary hurdles (connectivity, skills, language) and the paid price drops out of it, whereas paying for more capability depends on the ratio of price to income. Proposition 2 in Appendix A then predicts a steeper income gradient on the paid margin than on the margin of any use whenever prices are not localised, and the difference falls as localisation increases. If the paid option is attractive enough to draw users directly from non-use, the paid price affects both margins. The model conditions on common participation costs and service levels; it organises the evidence and generates predictions, but the public usage data do not observe paid conversion, so it is not fitted or tested below.

3.3 Conditional gains and unconditional gaps

Suppose that AI raises output in economy i by $\Delta Y_i/Y_i = s_L E_i a_i g_i$, where s_L is the labour share, E_i the share of labour tasks exposed to AI, a_i the adoption rate in exposed tasks and g_i the task-level productivity gain, as in Hulten-style accounting applied to AI (Hulten, 1978; Acemoglu and Restrepo, 2019; Acemoglu, 2025). For two economies (or groups) H and L with output per worker $y_H > y_L$, the relative gap moves from $R_0 = y_H/y_L$ to

$$\frac{R_1}{R_0} = \frac{1 + s_L E_H a_H g_H}{1 + s_L E_L a_L g_L}. \quad (2)$$

Proposition 1 (Access-conditional equalisation). *The relative gap narrows if and only if $E_L a_L g_L > E_H a_H g_H$. Equivalently, the adoption rate in L needed to avoid divergence is*

$$a_L^* = a_H \cdot \frac{E_H}{E_L} \cdot \frac{g_H}{g_L}.$$

Larger conditional gains for the less productive ($g_L > g_H$) are neither necessary nor sufficient for convergence.

Proof. From (2), $R_1 < R_0$ if and only if $E_L a_L g_L > E_H a_H g_H$; solving the equality for a_L gives a_L^* . If $g_L > g_H$ but $E_L a_L < E_H a_H g_H/g_L$ the gap widens, so $g_L > g_H$ is not sufficient; if $g_L \leq g_H$ but $E_L a_L$ is large enough the gap narrows, so it is not necessary. $\square$

The proposition separates two statements that are often conflated. “AI helps the less skilled more” is a statement about g conditional on use, whereas “AI narrows gaps” also depends on exposure E and adoption a , both of which rise with income. Evidence on g is itself mixed outside the settings cited above: in a randomised trial with Kenyan entrepreneurs, access to a GPT-4 business mentor raised the performance of initially strong performers by over 15% and lowered that of weak performers by nearly 10% (Otis et al., 2025). Rough proxies indicate the magnitudes involved. The share of employment in occupations exposed to generative AI is 34% in high-income and 11% in low-income economies (Gmyrek et al., 2025), and the shares of the working-age population using any generative-AI product are about 32% and 9% (Section 5.3; the latter includes region-imputed values). With these figures standing in for task exposure and within-task adoption, a_L^* is 1.00 if task-level gains are equal, 0.67 if they are 1.5 times larger in L and 0.50 if twice as large. The calculation gives orders of magnitude, not

estimates. Its point is that conditional equalisation does not aggregate into convergence unless access and exposure keep pace; I refer to the possibility that AI equalises conditional on use yet widens gaps unconditionally as the *AI access paradox*.

3.4 What the public usage indicators identify

The paid-adoption share of the model is not observed in the public data. Microsoft’s AI User Share is an estimated adoption rate for a set of tracked products. Anthropic’s AI Usage Index is a population-normalised measure of Claude conversation activity that pools free and paid accounts; it does not identify the number of paid users or their usage separately. The empirical comparison therefore concerns two observed diffusion indicators, not separately measured extensive and paid margins.

An accounting identity shows the distinction. In a common period and population, let N_c denote population, U_c any-AI users, C_c Claude users and M_c Claude conversations, with Claude users a subset of any-AI users. Define $D_c^* = U_c/N_c$, $s_c = C_c/U_c$ and $m_c = M_c/C_c$. For positive counts,

$$\frac{M_c}{N_c} = D_c^* s_c m_c, \quad AUI_c = K D_c^* s_c m_c, \quad K = \frac{N_W}{M_W}, \quad (3)$$

where W is the common reference universe used to normalise the index. Consequently, under aligned definitions,

$$\varepsilon_{AUI} = \varepsilon_{D^*} + \varepsilon_s + \varepsilon_m. \quad (4)$$

The same additive relation holds for ordinary least-squares log-income slopes estimated on an identical country sample. The country-invariant normalisation K affects the intercept rather than the cross-country slope.

A steeper conversation-activity gradient can thus reflect an income gradient in Claude’s share of AI users, in conversations per Claude user, or both. Neither component is synonymous with paid access: with π_c the paid share of Claude users and $m_{F,c}, m_{P,c}$ the mean conversation counts of free and paid users, $m_c = (1 - \pi_c)m_{F,c} + \pi_c m_{P,c}$, and the index identifies none of these terms. In practice the two sources also do not provide the harmonised counts of equation (3): Microsoft’s estimate incorporates telemetry coverage, scaling and shrinkage, while Anthropic uses sampled conversations, publication thresholds and release-specific product coverage, so the observed gradient difference can reflect measurement and selection as well as behaviour. The empirical analysis accordingly labels its outcomes any-product adoption and per-capita Claude activity, and reads their contrast as consistent with the mechanism of Section 3.2 rather than as a test of it. Testing paid upgrading would require country-level user counts by subscription status, usage quantities and effective prices, preferably around a plausibly exogenous change in access conditions.

4 Data

Table 2 lists sources and coverage, and Appendix B describes how the variables are constructed.

Income. The main denominator is GNI per capita (Atlas method) for 2025, available for 182 economies. For the remaining 19 I use the latest value from 2022–2024. Nigeria’s 2022 GNI per capita, for instance, has been revised from \$2,140 to \$2,940 in the current WDI vintage, while its 2025 value

Table 2: Data sources

Variable	Source and definition	Coverage
Income, groups	World Development Indicators (World Bank, 2026b): GNI per capita, Atlas method (current US\$) and PPP, household consumption; FY2027 income groups (World Bank, 2026a)	201 economies; 2025 for 182
Distribution	WDI income/consumption shares by quintile (latest survey 2010–2025)	163 economies
Local prices	Apple App Store product pages of ChatGPT, Claude and Gemini: in-app purchase lists and storefront currency, 27 Sep 2026; FX from open.er-api.com (cross-checked)	218 storefront codes; 170 with prices
Any-use diffusion	Microsoft AI User Share, H1 2025–Q2 2026 (Misra et al., 2025; Microsoft AI Economy Institute, 2026b)	147 economies; 111 country-specific
Claude activity	Anthropic AI Usage Index (AUI), Aug 2025–May 2026 (Anthropic, 2026a)	114–121 economies
Gemini usage rank	Google ATLAS v1.0 per-capita adoption quintile, Apr 2026 (Ischenko et al., 2026)	195 economies
Complements	Internet use, electricity, tertiary enrolment, urbanisation (WDI); Global Findex 2025 (Klapper et al., 2025); ITU ICT price baskets (International Telecommunication Union, 2025b)	142–217 economies
Exposure	Employment exposed to generative AI: 34% high-income, 11% low-income (Gmyrek et al., 2025)	Income groups

Notes: Income groups follow the World Bank classification for fiscal year 2027 (effective 1 July 2026), based on 2025 GNI per capita: low $\leq$ \$1,175; lower-middle \$1,176–4,635; upper-middle \$4,636–14,375; high $>$ \$14,375.

is \$1,360 after the naira’s depreciation. I use the official FY2027 groups throughout. For Argentina (\$14,650) and Türkiye (\$16,300) the 2025 values in the July 2026 WDI update exceed the high-income threshold, although both are classified as upper-middle income.

Local prices. For each of 218 two-letter storefront codes and each app, I retrieved the App Store product page and extracted the storefront currency and the list of in-app purchases with their displayed prices (the list’s title is localised, e.g. “In-App-Käufe”). Plans are identified by name. Where a plan appears with monthly and annual variants, the monthly price is the modal price after discarding variants priced at five or more times the lowest one. The archive retains parsed listing records rather than the original HTML; billing-period inference from names and price magnitudes is a matching heuristic, not independently confirmed SKU metadata. Prices are converted to dollars at market rates of 27 September 2026 from open.er-api.com, which agree with an independent source to within 1.6%. Requests redirected to the U.S. storefront are coded as no local listing observed (45 codes). Pages returning “not found” are coded as unavailable in the audit (ChatGPT: Belarus, China, Hong Kong SAR, Macao SAR, Russia and Venezuela; Claude: the same plus Afghanistan, Myanmar and Yemen; Gemini: Afghanistan, Belarus, China and Russia). App Store prices include value-added or goods-and-services taxes in almost all storefronts and exclude sales taxes in the United States and Canada. They can also exceed web prices because of platform commissions (Claude Max, for example, costs \$124.99 in the U.S. App Store against “from \$100” on the web). The audit therefore measures listed prices in one platform-specific payment channel. These are not the vendor’s web list prices, and they are not necessarily the amounts charged at checkout, which introductory offers or legacy price points can change.

Diffusion. The Microsoft AI User Share is the estimated share of people aged 15–64 who used a generative-AI product during the period, derived from anonymised Windows telemetry on visits to 19 AI services, scaled for device and operating-system market shares, mobile use and internet penetration, and shrunk toward the global average for countries whose users mostly opt out of telemetry (Misra et al., 2025). Two features of the series affect the analysis. First, economies with insufficient telemetry or populations below two million are pooled into regional aggregates: 36 of the 147 economies share identical series with their neighbours, and they coincide with the economies flagged as “region-imputed” in the technical paper. The pooled groups include 20 of the 23 low-income economies in the data (for example, Benin, Ghana, Niger, Nigeria and eight other West African economies all record 8.7%, 9.3%, 10.1% and 10.4% in the four periods). Country-specific estimates are therefore available for only three low-income economies (The Gambia, Rwanda and Syria). I use country-specific estimates as the main sample and show results with pooled values, and with each pool collapsed to one observation, as robustness checks. Second, Microsoft cautions that several widely used domestic tools, notably in China, are not yet covered (Microsoft AI Economy Institute, 2026b).

The Anthropic AI Usage Index for country c is the country’s share of Claude.ai conversations divided by its share of the world’s working-age population, from privacy-preserving samples of about one million conversations per window (Handa et al., 2025; Appel et al., 2025, 2026). Later releases include consumer Free, Pro and Max accounts without distinguishing their country-level contributions; those account types were already included in the November 2025 sample. The consumer series excludes API and enterprise traffic and is published only for countries meeting release-specific thresholds. I harmonise the index normalisation across releases, without making changing product coverage comparable (Appendix B). Google’s ATLAS reports a per-capita adoption quintile for Gemini products in April 2026 (Ischenko et al., 2026).

Computational workflow. Generative AI tools, ChatGPT and Codex (OpenAI) and Claude (Anthropic), assisted with code development, extraction review and drafting; the author verified the sources, analysis and text. The replication package (Appendix B) contains the frozen inputs, scripts and outputs. Re-running it reproduces every number in the paper, and independent reimplementations of the key regressions in plain NumPy match the reported estimates.

4.1 Estimands and inference

For plan k , the price regression is $\ln p_{ck} = \alpha_k + \beta_k \ln y_c + u_{ck}$, estimated by ordinary least squares with equal country weights. The coefficient is the cross-country association of listed consumer prices with income within the observed sample; it is not a demand elasticity, and it is not a firm’s localisation rule, because taxes, platform commissions and product differences all enter the listing. Because $\ln AAB_{ck} = \ln(1200) + \ln p_{ck} - \ln y_c$, the burden slope on the same sample is exactly $\beta_k - 1$, and a constant reference price imposes a slope of -1 by construction; regressing use on that burden would supply no price variation independent of income. The matched Go-to-Plus regression uses the ratio of two listings observed in the same storefront and then compares ratios across economies; it cancels the currency conversion common to both plans but supplies no within-country variation over time.

Usage regressions take the form $\ln U_c = \alpha + \eta \ln y_c + X'_c \gamma + e_c$. Common-sample comparisons use the same countries and income observations for both outcomes. HC3 standard errors describe regression

Table 3: AI affordability burden by income group: annual cost as % of GNI per capita (medians)

Income group	<i>N</i>	\$8	\$20	\$100	\$200	HH cons.	PPP indexed
High income	75	0.26	0.65	3.24	6.48	1.18	0.40
Upper-middle income	58	1.23	3.09	15.43	30.87	4.53	1.20
Lower-middle income	47	3.49	8.73	43.64	87.27	11.85	3.38
Low income	21	11.57	28.92	144.58	289.16	35.57	10.67

Notes: Annual cost as % of Atlas GNI per capita. 2025 income for 182 economies; latest 2022–2024 for 19. The last two columns divide the \$20 price by household final consumption per capita and by GNI per capita at PPP, which equals the burden if prices were indexed to local price levels (Remark 1).

uncertainty conditional on the published indicators; currency-clustered errors are reported for the price regressions, and country-clustered and bootstrap intervals for the difference between usage gradients. None of these incorporates the estimation uncertainty inside the providers’ own indicators or selection into publication. Population weighting is reported as a sensitivity because it describes a different estimation.

The audit date, income year, diffusion window and survey year are distinct, and the design neither links contemporaneous price changes to later use nor estimates individual willingness to pay.

5 Results

5.1 Burdens

Table 3 reports median burdens by income group for common reference prices from \$8 to \$200 a month. At \$20 the median burden is 0.65% of per-capita income in high-income economies, 3.09% in upper-middle-, 8.73% in lower-middle- and 28.9% in low-income economies, a 45-fold difference between the first and last groups. The \$20 tier meets the 2% benchmark in every high-income economy, in 22% of upper-middle-income economies and in none of the lower-middle- or low-income economies. The 113 of 201 economies that exceed it are home to 60% of the working-age population of the economies covered (which account for 98.6% of the world’s), and 45% of the covered working-age population lives where the burden exceeds 5%. The \$8 entry tier meets the benchmark in 95% of upper-middle-income economies but in no lower-middle- or low-income economy. At a \$100 monthly reference fee the median low-income burden exceeds per-capita income itself.

Relative to household consumption rather than GNI, burdens are 23–81% higher (1.18%, 4.53%, 11.9% and 35.6%). Relative to the ITU’s entry-level mobile-broadband basket, the \$20 plan costs 1.7 times as much in the median high-income economy but 6.4 times as much in the median low-income economy (Table 4). In the poorest countries the AI subscription thus costs several times as much as the internet connection it runs on. The PPP-indexed counterfactual in the last column would lower the median low-income burden from 28.9% to 10.7%, which is still five times the benchmark.

Burdens at quintile means give a fuller picture (Figure 1, Table 4). For the poorest fifth of households, a \$20 plan costs a median 1.6% of mean income in high-income economies but 33% in lower-middle- and 73% in low-income economies. The plan fails the 2% benchmark even for the richest fifth in every low-income economy with survey data (19) and in 98% of lower-middle-income economies; in these two groups the median burden for the richest quintile is 13% and 4.4%. Households at the top of the distribution in these economies, where the professionals most likely to use AI at work are con-

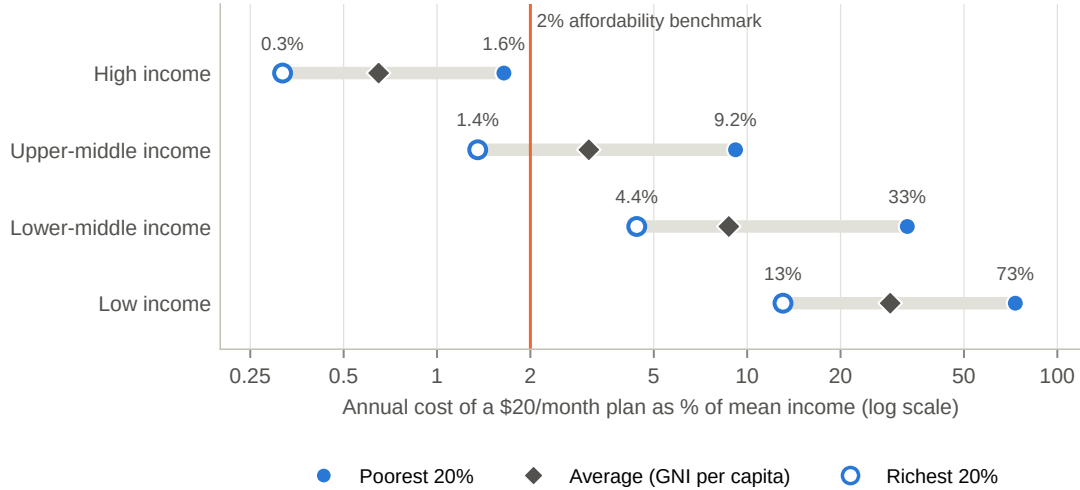


Figure 1: Burden of a \$20 plan across the income distribution. Medians across economies by income group of the annual cost of a \$20/month plan as a share of the mean income of the poorest quintile, of the population (GNI per capita), and of the richest quintile. Quintile incomes are approximated as GNI per capita times the survey income share relative to 20%. The vertical line marks the 2% broadband affordability benchmark.

Table 4: Burden of a \$20 plan by income quintile, and comparison with mobile broadband (medians)

Income group	N	\$20 plan, % of quintile mean income			Mobile broadband 2 GB, % GNI p.c.	AI plan / broadband
		Poorest 20%	Middle 20%	Richest 20%		
High income	47	1.6	0.7	0.3	0.43	1.7
Upper-middle income	53	9.2	3.9	1.4	1.16	2.6
Lower-middle income	44	32.8	13.1	4.4	2.31	5.1
Low income	19	73.2	36.1	13.0	8.22	6.4

Notes: Quintile shares from the latest household survey in WDI (median survey year 2021–2023). Mobile broadband: ITU data-only basket with at least 2 GB per month, the selected 2024 basket, as a share of monthly GNI per capita (International Telecommunication Union, 2025b). The last column is the median of \$20 divided by the 2024 broadband basket price in current dollars; it does not divide burdens with different income vintages. Broadband and survey columns use their available samples, which need not coincide.

centrated, therefore face burdens several times those of the poorest fifth of households in high-income economies.

Sensitivity to the benchmark and the income vintage. The 2% benchmark is borrowed from broadband policy rather than derived for AI, so Table 5 reports other cutoffs. At a \$20 reference price, the share of the covered working-age population living in economies above the cutoff is 85.6% at 1%, 60.1% at 2%, 44.6% at 5% and 16.7% at 10%. Restricting to the 182 economies with 2025 GNI gives 60.0% at 2%, with median burdens of 0.67%, 3.09%, 8.87% and 28.9% across the four groups, so the 19 older income observations do not drive the pattern.

5.2 Local prices: how much do vendors localise?

Burdens computed at U.S. list prices would overstate the gradient if vendors charged less in poorer markets. The audit asks how much the observed price schedule offsets the mechanical burden slope of minus one. Figure 2 plots the local App Store price of ChatGPT Plus and ChatGPT Go against per-capita income, and Table 6 reports, for nine plans, median local prices by income group and the slope $\hat{\beta}$ from

Table 5: Share of economies and population above alternative benchmarks, \$20 plan

Cutoff	Economies above	Share of economies	Share of covered population
1%	140	69.7%	85.6%
2%	113	56.2%	60.1%
5%	71	35.3%	44.6%
10%	42	20.9%	16.7%

Notes: 201 economies. Population is ages 15–64. The 2% row is the broadband benchmark used in the text; the other rows are alternative cutoffs.

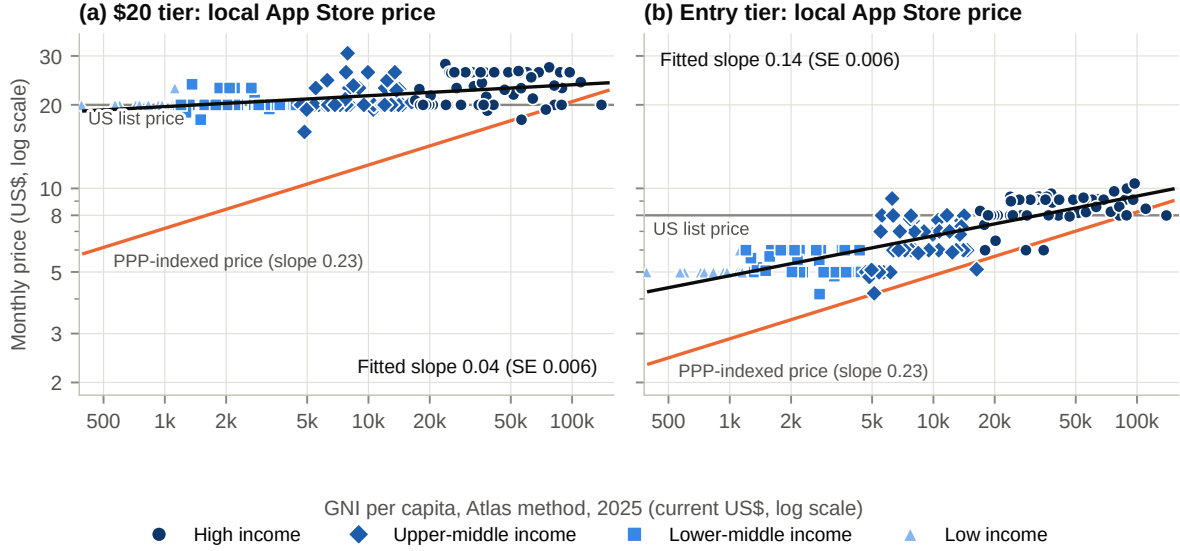


Figure 2: Local App Store prices of ChatGPT Plus and ChatGPT Go against income. Tax-inclusive monthly prices in each national App Store storefront, converted to dollars at market exchange rates on 27 September 2026. Black lines: fitted log-log regressions. Orange lines: prices indexed to local price levels, anchored at the U.S. price, with the slope (0.23) on income of the Atlas-to-PPP income ratio, a price-level proxy. Grey lines: U.S. list prices.

regressing log dollar price on log income.

The standard and high-usage tiers are priced almost uniformly. For ChatGPT Plus, $\hat{\beta} = 0.039$ (s.e. 0.006), and the estimates for Claude Pro and Google AI Pro are 0.026 and 0.037. The median local price of ChatGPT Plus is \$19.99 in upper-middle-, lower-middle- and low-income storefronts alike, and it is higher, at \$22.99, in the median high-income storefront. Because App Store prices include value-added taxes outside the United States and Canada, the small positive slopes may reflect taxes rather than price discrimination. Only OpenAI’s entry tier is localised to a meaningful degree: ChatGPT Go has $\hat{\beta} = 0.143$, with median prices of \$9.05, \$5.99, \$5.34 and \$4.99 across the four groups, and lows of \$4.16 in India and \$4.19 in Indonesia. On the same 164-economy sample the price-level proxy has an income slope of 0.243, so Go’s slope is about 60% of the slope implied by indexing to local price levels and one-seventh of proportional pricing. Google’s entry tier, AI Plus, is flat at \$4.99 in each of the three non-high-income groups.

With β near zero, the burden elasticity $\beta - 1$ is -0.96 for the standard tier: local pricing passes almost the whole income gradient through to burdens. As Table 7 shows, every storefront in a low-income economy, and 86% of those in lower-middle-income economies, prices in U.S. dollars. In 36% of low-income and 21% of lower-middle-income economies there is no local storefront at all, so residents must pay through a foreign account or on the web, typically with an international card. The cheapest

Table 6: Local prices and localisation slopes by plan

Plan	U.S. price	Median local price (US\$)				Income slope		N
		High	Upper-mid.	Lower-mid.	Low	$\hat{\beta}$	(s.e.)	
ChatGPT Go	8.00	9.05	5.99	5.34	4.99	0.143	(0.006)	164
Google AI Plus	4.99	5.58	4.99	4.99	4.99	0.026	(0.004)	160
ChatGPT Plus	19.99	22.99	19.99	19.99	19.99	0.039	(0.006)	164
Claude Pro	20.00	23.94	20.00	20.00	20.00	0.026	(0.006)	162
Google AI Pro	19.99	22.77	19.99	19.99	19.99	0.037	(0.005)	165
ChatGPT Pro 5x	100.00	114.53	100.00	100.00	100.00	0.019	(0.004)	164
Claude Max 5x	124.99	149.99	124.99	129.99	124.99	0.035	(0.006)	162
ChatGPT Pro 20x	200.00	234.06	200.00	200.00	200.00	0.033	(0.006)	164
Google AI Ultra	199.99	205.60	199.99	199.99	199.99	0.030	(0.006)	95

Notes: Monthly App Store prices, 27 September 2026, tax-inclusive except in the U.S. and Canada. $\hat{\beta}$: OLS slope of $\ln(\text{local dollar price})$ on $\ln(\text{GNI per capita})$, heteroskedasticity-robust (HC3) standard errors. For reference, the slope on log income of the log price-level proxy (Atlas over PPP GNI) is 0.228 (s.e. 0.014), and that of the GDP price-level ratio 0.224. Proportional pricing would imply a slope of 1.

Table 7: Storefront availability, currency and the cheapest locally listed paid tier

Income group	N	ChatGPT storefront (% of econ.)			USD priced (%)	Cheapest paid tier		Plus burden
		Local	None	Not offered		US\$	Burden	
High income	86	73	23	3	35	5.65	0.17	0.75
Upper-middle income	59	86	10	3	69	4.99	0.81	3.27
Lower-middle income	47	77	21	2	86	4.99	2.15	8.89
Low income	25	64	36	0	100	4.99	7.21	28.90

Notes: “No store”: a redirect to the U.S. store was observed. “Not offered”: the request returned no app page. These are audit retrieval outcomes, not complete service-availability determinations. Cheapest paid tier: the lower of ChatGPT Go and Google AI Plus in local storefronts. Burdens are medians of $12p/y \times 100$ at local prices.

paid tier listed locally costs \$4.99 in the median low-income economy, a burden of 7.2%, and it meets the 2% benchmark nowhere in that group; in lower-middle-income economies it does so in 38% of cases.

The pattern is consistent with the logic of menu design. Localising the \$20 tier invites arbitrage by high-valuation users through foreign accounts, whereas localising a capability-limited entry tier does not, because high-valuation users self-select into the higher tier (Mussa and Rosen, 1978). Where vendors price-discriminate across countries, they sell degraded quality cheaply but barely localise the prices of frontier tiers. By Proposition 2, price should then matter most on the paid margin, which remains unlocalised. Taxes, platform rules and currency schedules also enter the listings, so the slopes are associations in a price menu rather than firms’ discrimination parameters, and the audit cannot establish whether a lower-allowance tier suffices for any particular task.

Robustness: currency dependence and a matched-plan comparison. Prices in the same currency share exchange-rate and platform-pricing shocks, and the currency clusters are highly unbalanced: the Plus sample has 102 USD listings, 25 EUR listings and 37 singleton currencies. Currency-clustered standard errors are accordingly larger than HC3 errors (Table 8). For ChatGPT Plus the clustered 95% interval is $[-0.0002, 0.079]$, so the small positive standard-tier slope is not distinguishable from zero under that dependence assumption, which only strengthens the uniformity finding; the Go slope’s clustered interval, $[0.114, 0.173]$, excludes zero. A matched comparison removes currency conversion altogether.

Within each storefront, the log Go-to-Plus price ratio has an income slope of 0.104 (currency-clustered s.e. 0.007; 164 economies), which on a common sample is exactly the difference between the Go and Plus slopes. Its value is not new information but a check: because the ratio cancels the exchange-rate conversion and any proportional tax common to both plans, the entry-tier gradient is not an artefact of currency conversion. It remains 0.113 within the 102 USD-priced storefronts, 0.112 without them and 0.125 without USD and EUR (Table 9 and Figure 3), although no low-income economy remains once USD listings are excluded, so the exclusion samples speak to middle- and high-income markets. Restricting the separate regressions to economies with 2025 income data gives slopes of 0.149 for Go and 0.046 for Plus.

Table 8: Price slopes under alternative standard-error assumptions

Plan or ratio	Slope	HC3 SE	Currency SE	Cluster 95% CI	<i>N</i>
ChatGPT Go	0.143	0.006	0.014	[0.114, 0.173]	164
ChatGPT Plus	0.039	0.006	0.020	[-0.0002, 0.0790]	164
Claude Pro	0.026	0.006	0.014	[-0.002, 0.055]	162
Google AI Pro	0.037	0.005	0.016	[0.004, 0.070]	165
Go / Plus ratio	0.104	0.006	0.007	[0.089, 0.119]	164

Notes: All outcomes are in logs. Currency-clustered CR1 errors with t_{G-1} intervals; 39 clusters except Google AI Pro (40). Currency clusters are highly unbalanced (102 USD, 25 EUR, 37 singletons). The ratio uses the two plans as listed in the same storefront.

Plan matching. Monthly prices are selected from the parsed in-app purchase lists by name and price magnitude (Section 4). Re-implementing those rules identifies a single surviving candidate price in every ChatGPT Go, ChatGPT Plus and Claude Pro storefront-plan cell, and for the two ChatGPT plans annual-price exclusion cutoffs of three, five and eight times the lowest listed price select the same listings. Google AI Pro has three cells with two candidate prices; excluding them changes its slope from 0.0370 to 0.0374. These checks show that the results do not depend on the particular selection rule. They do not verify billing-period metadata or checkout eligibility, because the archive stores parsed records rather than the original pages (Section 6.4).

Table 9: The Go-to-Plus price ratio by currency sample

Sample	Slope	HC3 SE	HC3 95% interval	<i>N</i>
All matched storefronts	0.104	0.006	[0.093, 0.115]	164
USD only	0.113	0.008	[0.098, 0.128]	102
Exclude USD	0.112	0.020	[0.073, 0.151]	62
Exclude USD and EUR	0.125	0.023	[0.079, 0.171]	37

Notes: OLS of the log Go-to-Plus listing ratio on log Atlas GNI per capita. Intervals use HC3 standard errors with a normal reference. Each restriction changes the sample; no low-income economy remains in the non-USD samples.

5.3 Any use: the extensive margin

Median AI user shares in the second quarter of 2026 were 32.3% in high-income, 18.8% in upper-middle-, 12.0% in lower-middle- and 7.6% in low-income economies with country-specific estimates (Figure 4a). Table 10 reports the cross-country gradient. On country-specific estimates, a one-log-point increase in income is associated with 8.3 percentage points higher use, and the elasticity is 0.370 (s.e.

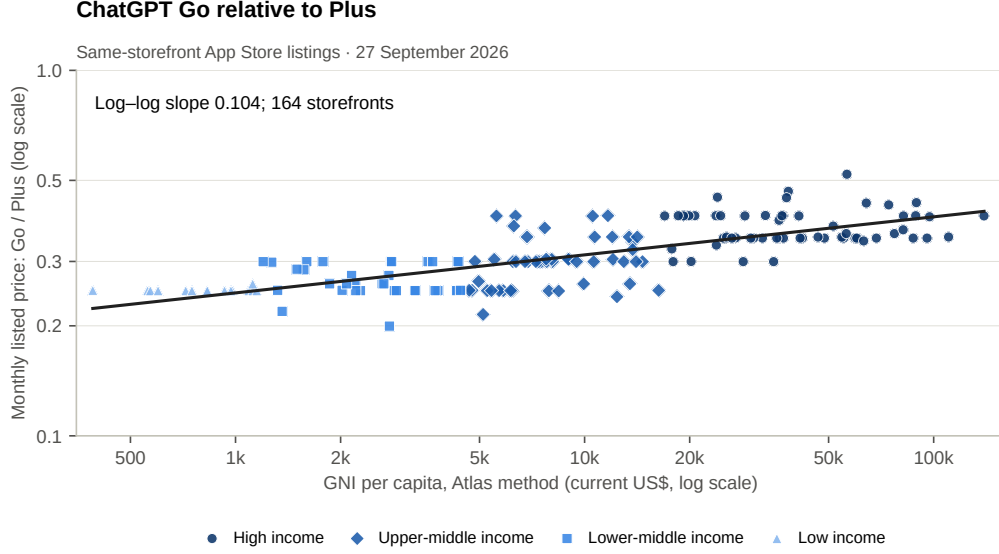


Figure 3: The ChatGPT Go-to-Plus price ratio within storefronts. Both plans as listed in the same storefront on 27 September 2026; each dot is a storefront. The line is the log–log fit.

Table 10: Income gradient of generative-AI use (Microsoft AI User Share, Q2 2026)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Level (pp)	Log user share					
ln GNI per capita	8.326 (0.655)	0.370 (0.021)	0.336 (0.015)	0.367 (0.019)	0.360 (0.024)	0.357 (0.062)	0.367 (0.067)
Controls	No	No	No	No	No	Yes	Yes
Region fixed effects	No	No	No	No	No	No	Yes
Sample	CS	CS	All	Coll.	CS [†]	CS [†]	CS [†]
Observations	109	109	142	115	97	97	97
R^2	0.67	0.71	0.72	0.73	0.69	0.73	0.74

Notes: OLS with HC3 standard errors in parentheses. CS: country-specific estimates only; All: including region-imputed values; Coll.: each region-imputed pool collapsed to one observation with population-weighted log income. [†]Common sample with non-missing controls (internet use, tertiary enrolment, urban share, population 65+, share making digital online merchant payments). Coefficients on controls in Appendix Table 14.

0.021; $R^2 = 0.71$). Including pooled regional values lowers the elasticity to 0.336; collapsing each pool to one observation restores it (0.367). The elasticity is unchanged by controls for internet use, tertiary enrolment, urbanisation, age structure, digital payments and region fixed effects (0.355–0.367). Conditional on income, none of the controls is individually significant except urbanisation in columns (3)–(4) of Appendix Table 14 ($p \approx 0.04$), which loses significance once region fixed effects are added.

About half of the gradient coincides with connectivity. Because use requires connection, the user share can be written as the product of the internet-use rate and the ratio of AI users to internet users; regressing each log component on log income decomposes the elasticity exactly, with connectivity contributing 0.182 and the ratio 0.189, so the connectivity component accounts for 49% of the gradient (Appendix C). The ratio of AI users to internet users is about 23% in the upper-middle-, lower-middle- and low-income groups alike, against 37.5% in high-income economies, and its slope on log income across non-high-income economies is -0.03 (s.e. 0.06) even though the burden of a paid plan rises nine-fold across those groups. The decomposition is an accounting identity rather than a causal share, and

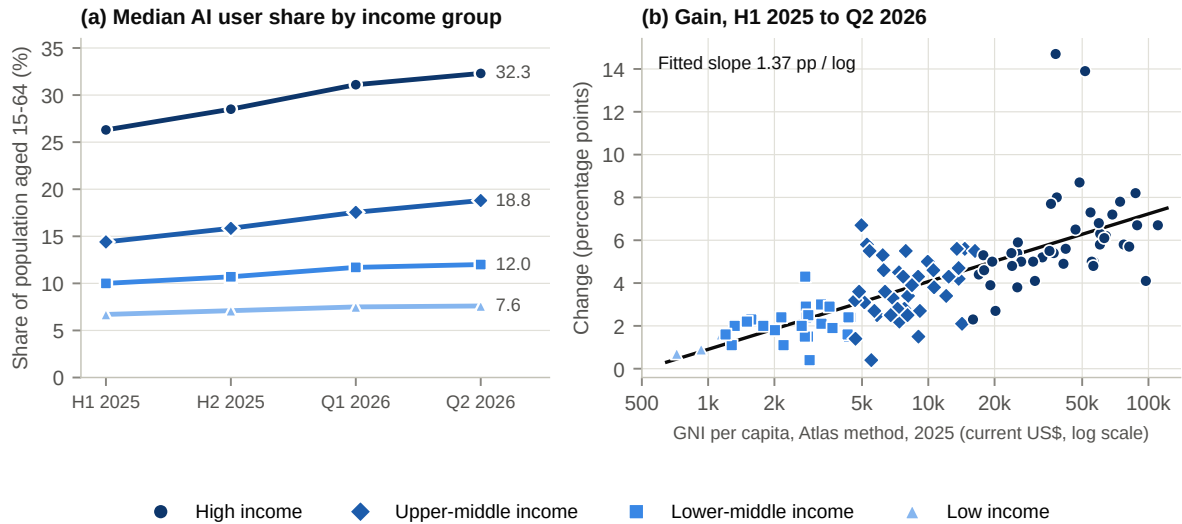


Figure 4: Diffusion dynamics. (a) Median AI user share by income group, country-specific Microsoft estimates (low-income: three economies). (b) Change in the AI user share between H1 2025 and Q2 2026 against GNI per capita, with the fitted log-linear slope.

the internet-use series has a different age base and already enters Microsoft’s estimator. Still, a binding price constraint on the extensive margin would make the ratio fall as the burden rises, and across the three groups it does not; the pattern is what Proposition 2 predicts when a free tier exists, although the flatness itself is not implied by the model and may partly reflect measurement.

Changes in relative and absolute dispersion. Between the first half of 2025 and the second quarter of 2026 the median economy’s user share rose by 5.5 points in the high-income group, 3.7 in the upper-middle-, 2.0 in the lower-middle- and 0.9 in the low-income group (Figure 4). Proportional growth was similar in the three higher groups (21–26%), and lagging economies grew slightly faster in logs: regressing log growth on the initial log share gives -0.026 (s.e. 0.011), and the cross-country standard deviation of log shares fell from 0.564 to 0.553. In percentage points the opposite holds: each log point of income is associated with a 1.37-point larger gain (s.e. 0.13), the standard deviation rose from 10.9 to 12.8 points, and the gap between the median high- and lower-middle-income economy widened from 16.3 to 20.3 points. This reconciles reports of faster proportional growth in lower-income countries (Chatterji et al., 2025) and of a widening divide (Microsoft AI Economy Institute, 2026a; Liu et al., 2025), which are two readings of an early-phase diffusion curve, although four vendor-estimated windows cannot establish its shape and measurement error in the initial level also affects growth-on-initial-level regressions.

5.4 Per-capita Claude activity: an imperfect view of the paid margin

Anthropic’s own report documents an income gradient of Claude activity near 0.7 using GDP per working-age capita (Appel et al., 2025). The contribution here is the comparison with a different public indicator on a common sample, not the gradient itself.

Figure 5 plots both measures, as log deviations from their cross-country means, against income for the 96 economies covered by both: 42 high-, 33 upper-middle-, 20 lower-middle- and one low-income economy (Rwanda). The comparison therefore describes the gradient from lower-middle to high income and says little about low-income economies. Per-capita Claude activity rises twice as steeply

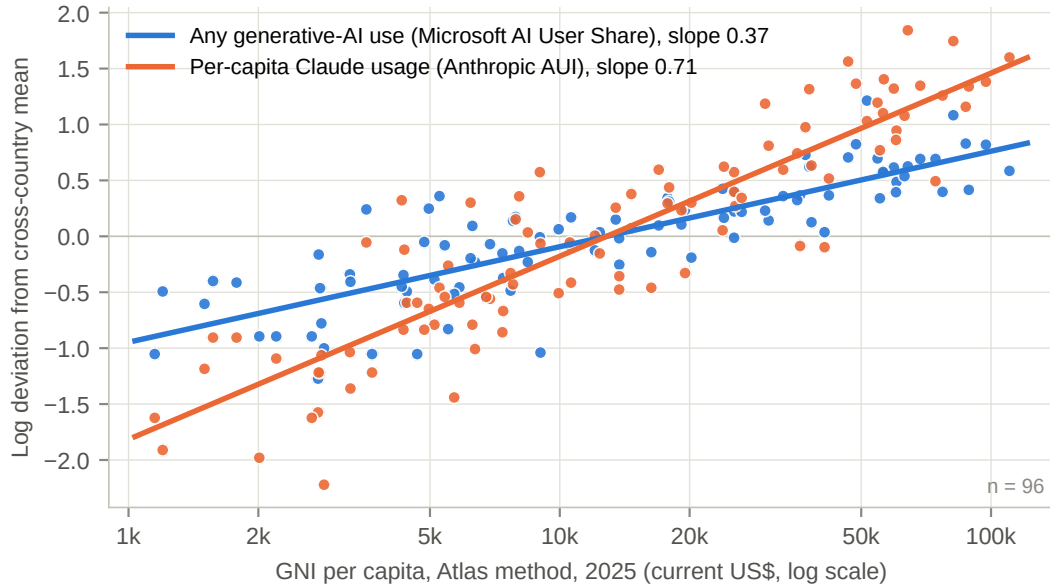


Figure 5: Two margins of AI diffusion. Log deviation from the cross-country mean of the Microsoft AI User Share (any generative-AI use, Q2 2026, country-specific estimates) and of the Anthropic AI Usage Index (per-capita Claude use, May 2026) against GNI per capita, for the 96 economies covered by both. Lines are OLS fits; slopes are income elasticities.

with income: its elasticity is 0.711 (s.e. 0.036) against 0.371 (s.e. 0.026) for any use, a difference of 0.340 (clustered s.e. 0.034; bootstrap 95% interval 0.27–0.40). Table 11 shows that the gap is not an artefact of functional form or sample. It persists when the user share is expressed in log-odds to allow for ceiling effects (0.50 versus 0.71), when high-income economies are excluded (0.33 versus 0.69), conditional on internet use, tertiary enrolment and urbanisation (0.31 versus 0.55), in earlier measurement windows (0.38 versus 0.62), and when pooled regional values are included (0.36 versus 0.73).

The measured gradient also steepened, though not continuously. On the 111 economies observed in all five windows, the elasticity was 0.68 in August and November 2025, 0.74 in February 2026, 0.77 in April and 0.73 in May 2026. The increase from August 2025 to February 2026 (0.064, clustered s.e. 0.018) is significant, whereas the change from February to May 2026 is not (–0.006, s.e. 0.016). Anthropic’s releases changed over this period, with later releases adding Cowork sessions and monthly windows, so the series describes measured Claude use rather than a strictly like-for-like quantity; estimates on all economies meeting Anthropic’s thresholds in each window are similar (Appendix Table 15), and the steepening is in line with Anthropic’s own finding that low-adoption countries fell slightly further behind (Massenkoff et al., 2026). The composition of use differs as well: the share of Claude conversations devoted to coursework falls by 4.7 points per log point of income, from 35% in the median low-income economy (five economies) to 17% in the median high-income economy. This gradient is consistent with evidence that learning is a leading entry point in lower-income regions (Microsoft AI Economy Institute, 2026b; Daepf and Slaughter, 2026) and with Anthropic’s observation that coursework may be better suited to free use (Appel et al., 2026). A third source, Google’s per-capita adoption quintile for Gemini, is also positively but less tightly related to income (Spearman correlation 0.46 across 195 economies), although a rank measure and different product coverage prevent direct elasticity comparison.

Table 11: Income elasticities of any use and of per-capita Claude activity

Specification	Any use (MS)	Claude activity (AUI)	N
Baseline: country-specific MS, Q2 2026 vs AUI, May 2026	0.371 (0.026)	0.711 (0.036)	96
MS in log-odds (ceiling check)	0.496 (0.037)	0.711 (0.036)	96
Excluding high-income economies	0.334 (0.060)	0.688 (0.089)	54
Conditional on internet use, tertiary enrolment, urbanisation	0.315 (0.050)	0.547 (0.064)	92
Earlier windows: MS H2 2025 vs AUI Aug 2025	0.380 (0.028)	0.619 (0.047)	93
Including region-imputed MS values	0.358 (0.020)	0.733 (0.030)	108
Difference in baseline elasticities (AUI – MS)	0.340 (0.034); bootstrap 95% CI [0.27, 0.40]		96

Notes: Cells are OLS slopes on log GNI per capita in common samples; outcomes are in logs except the explicitly labelled Microsoft log-odds row, which is not the same elasticity. Parentheses give HC3 errors except for the stacked difference row. MS: Microsoft AI User Share; AUI: Anthropic AI Usage Index. The difference test stacks both outcomes with an outcome-specific intercept and slope, clustering by economy. The bootstrap resamples economies 2,000 times. The main sample comprises 42 high-, 33 upper-middle-, 20 lower-middle- and one low-income economy. Excluding high-income economies leaves 54.

Direct estimation of the gradient difference. On a common sample, the difference of the two log-income slopes equals the slope in a regression of $\ln(AUI_c/D_c)$ on log income, with or without an identical set of controls. The baseline difference is 0.340 (HC3 s.e. 0.034); it is 0.356 with 2025 GNI only, 0.266 with region fixed effects and 0.232 with internet, education and urbanisation controls (Table 12). Leaving out each country in turn gives 0.329–0.351, and working-age-population weighting gives 0.476, a different estimand dominated by large countries. None of these calculations separates the vendor-share and conversations-per-user components of equation (3).

Table 12: Sensitivity of the difference between the two income elasticities

Specification	Difference	HC3 SE	95% interval	N
Baseline	0.340	0.034	[0.273, 0.407]	96
2025 GNI only	0.356	0.034	[0.289, 0.423]	92
Non-high-income	0.354	0.085	[0.187, 0.521]	54
Region fixed effects	0.266	0.050	[0.169, 0.363]	96
Internet, education, urbanisation	0.232	0.068	[0.099, 0.365]	92
Population weighted	0.476	0.086	[0.308, 0.644]	96

Notes: Dependent variable is $\ln(AUI/D)$; baseline uses 96 common economies. Intervals use the normal approximation. Population weighting changes the estimand.

These patterns are what Proposition 2 predicts when the paid hurdle exceeds the free hurdle and localisation is weak ($\hat{\beta} \approx 0.04$ at the \$20 tier). They do not test it, because paid uptake is unobserved: vendor selection, conversation frequency, product coverage, work composition and differential measurement error can generate the same ordering of slopes, and the changing coverage of the Anthropic releases means that the steepening is not a clean longitudinal estimate of behavioural divergence. Section 6 weighs these alternatives.

6 Discussion

6.1 What the evidence does and does not show

The evidence supports three statements. Listed prices for the standard tier are nearly identical across countries, and the only material localisation is of a degraded entry tier. A uniform dollar price therefore imposes burdens that differ by a factor of about 45 across income groups, a gradient that survives alternative denominators, benchmarks and income vintages. And the two public usage indicators differ in their income gradients: any use is moderately income-graded, with about half of its gradient coinciding with connectivity and no decline in the ratio of AI users to internet users as the burden of a paid plan rises, whereas per-capita Claude activity is twice as income-graded and became more so between late 2025 and early 2026.

The evidence does not establish that prices cause the steeper Claude gradient. The free-tier model of Section 3.2 predicts the observed ordering, but so could several alternatives. Language: frontier assistants perform best in high-resource languages, and English-language interactions are over-represented where local languages are poorly supported (Daepp and Slaughter, 2026). Product and brand: Claude’s share among AI users may vary with awareness, occupation and competing products, and it is less known in many lower-income markets than ChatGPT, Gemini or Meta’s assistant. Work composition: Claude’s usage skews toward software and professional tasks, whose share of employment rises with income (Gmyrek et al., 2025). Measurement: Microsoft’s estimator shrinks noisy countries toward the global average and floors device-scaling factors, which compresses cross-country variation, while Anthropic’s publication thresholds select countries on conversation volume, and the index pools free and paid accounts, so conversations per user may reflect work practices rather than plan status. The robustness checks address some of these concerns, since the gap persists without high-income economies and conditional on education, but the only credible route to a price effect is variation in prices that is not a function of income.

6.2 A feasible evaluation agenda

Such variation now exists. ChatGPT Go was rolled out to India and then to 170 further countries on a staggered schedule (OpenAI, 2026d); Google introduced and repriced AI Plus and AI Ultra in 2025–2026 (Google, 2026); OpenAI changed the allowances and eligibility of its Pro tiers in September 2026 (Section 2.1); several markets saw temporary premium tiers bundled with mobile subscriptions; and App Store price schedules change with exchange rates and taxes. Combined with the quarterly Microsoft series and the monthly Anthropic series, these events support difference-in-differences and event-study designs that compare economies before and after a localised tier becomes available.

The requirements are demanding. Outcomes must be measured at the level of the intervention: effective prices, paid conversion, active users, use per user and cancellations for the provider and plan concerned, on stable population bases and product coverage. The timing of tier introductions may respond to market growth and coincide with marketing, bundles or model upgrades, so an event study needs sufficient pre-treatment observations, plausible comparison groups, checks for anticipation and spillovers, and estimators that accommodate heterogeneous effects across rollout cohorts (Callaway and Sant’Anna, 2021). A response of Claude usage to a ChatGPT price change measures a cross-provider effect, not ChatGPT’s own-price elasticity of paid adoption, and currency or tax changes also move

incomes and other purchases, so they are not automatically valid instruments. Randomised discounts or institutional-access offers, with free-tier substitution and employer-provided access recorded separately, would identify effects on take-up most cleanly. Vendors could make all of these designs far more precise by publishing country-level paid-subscription counts and effective prices.

6.3 Policy implications

The results bear on four policy questions, ordered by the margin they concern.

Connectivity as a precondition, for the extensive margin. Connection, devices and electricity are preconditions for any use, and half of the use gradient coincides with the connectivity gradient. The flat ratio of AI users to internet users below the high-income group is consistent with the view that access and complements, rather than price, limit whether the connected use AI at all (World Bank, 2026c; International Telecommunication Union, 2025a), although it does not demonstrate this, and compute and data infrastructure add a further layer (Lehdonvirta et al., 2024). Evidence from fast internet in Africa suggests that such investments raise employment and productivity in their own right (Hjort and Poulsen, 2019).

Localise frontier access, not only degraded tiers, for the paid margin. The welfare effects of third-degree price discrimination are ambiguous in general, but they tend to be positive when discrimination opens markets that uniform pricing would leave unserved (Varian, 1985; Aguirre et al., 2010), as may be the case for frontier tiers whose list price exceeds a quarter of average income. Local-currency store-fronts, carrier billing and time- or usage-limited frontier access at locally indexed prices are practical steps; the arbitrage risk that keeps the \$20 tier uniform can be contained with usage caps rather than by withholding frontier models. Whether a lower-allowance tier suffices for a given task is not something the audit can settle.

Institutional access. Evidence on AI’s productivity effects comes overwhelmingly from employer-provided tools rather than individual subscriptions (Brynjolfsson et al., 2025; Cui et al., 2024). Schools, universities, public agencies and employers can decouple access from household income; a World Bank trial in Nigeria found sizeable learning gains from a supervised GPT-4 tutoring programme (De Simone et al., 2025). Research on subsidies for new health products suggests that free trials can raise long-run adoption through learning rather than crowding it out (Dupas, 2014), while even small prices can sharply reduce take-up among the poor (Cohen and Dupas, 2010).

Transparency. Research on AI access is constrained by the absence of public data on paid subscriptions, effective prices and usage limits by country. Machine-readable disclosure of dated country-level prices, taxes, supported payment methods, plan limits and product definitions, and usage statistics that distinguish free from paid accounts and users from use per user, would allow the designs above to be implemented. Appendix F lists the fields such a record should contain.

6.4 Limitations

First, burdens are normalised prices rather than expenditures, and quintile incomes are approximations based on survey shares that mix income and consumption concepts; a person living in an economy with a high national-average ratio is not necessarily unable to afford a subscription. Second, the App Store audit observes listed prices in one channel on one day. Listed prices can differ from web prices and from amounts charged at checkout, they include platform commissions and taxes that the regressions do

not separate, and the archive stores parsed listing records rather than the original pages, so the monthly billing period of each listing is inferred from names and price magnitudes rather than confirmed from product metadata. Promotions and bundles are not captured. Third, the Microsoft series is a modelled estimate built on desktop telemetry, and country-specific values are unavailable for most low-income economies; results for that group rest on three economies or on regional pools, and the two-indicator comparison includes a single low-income economy. Fourth, the Anthropic index covers one vendor's consumer product and does not distinguish paid from free users, so it proxies the paid margin imperfectly, and its coverage changed across releases. Fifth, all regressions are cross-sectional and descriptive, and their standard errors omit the uncertainty inside the providers' estimators. Sixth, the framework holds preferences, dispersion and task value fixed, omits competing providers, employer access and welfare, and is not calibrated to the usage indicators; the convergence arithmetic uses occupational exposure and population-wide use as proxies for task exposure and within-task adoption.

7 Conclusion

Generative AI is priced as a global product and experienced as a local one. The standard \$20 plan is listed at almost the same price everywhere, so it costs less than 1% of average income in the median rich country and more than a quarter in the median poor one, and it fails the broadband affordability benchmark for most of the world's working-age population and, in low-income economies, even for the richest fifth. Only OpenAI's entry tier is partly localised, and its localisation reaches about 60% of what indexing to local price levels would imply.

The public usage evidence is consistent with free tiers having taken the price out of the decision whether to use AI at all: on that margin, about half of the income gradient coincides with connectivity, and the ratio of AI users to internet users does not fall as the burden of a paid plan rises. How much frontier capability people use, by contrast, remains tied to income; per-capita Claude activity has a gradient twice as steep, and it steepened between August 2025 and February 2026. Cross-country data cannot say whether price is what ties frontier use to income rather than language, product reach or the composition of work. The staggered rollout of localised tiers and the recent changes in allowances now make that question answerable, provided that prices, entitlements and paid use are recorded at the level at which they change.

The conditional evidence that AI helps less-experienced workers most is real, but conditional equalisation does not aggregate into convergence unless access and exposure keep pace, and on current numbers they do not. Whether they will depends more on who can access or afford frontier models, for how much use and how soon after release, than on how capable the models become.

Declarations

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work the author used ChatGPT and Codex (OpenAI) and Claude (Anthropic) to assist with literature search, coding and data processing, drafting and translation, and to check the methods and the replication package. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Data availability. All data are publicly available. The replication package (raw extracts, code and outputs) is archived in a public repository; the link is withheld for anonymised review and is given on the title page. Appendix B describes its contents.

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A A conditional framework for free and paid access

Consider a connected individual choosing between no use, a free tier and a paid tier. Let $v > 0$ denote the individual's monthly monetary valuation of one unit of effective AI service. For example, $v = \theta w$, where w is monthly income and θ captures the relevance of AI to the individual's activities. Let $0 < q_F < q_P$ describe the tiers' service levels, $p_c > 0$ the monthly paid price, and $\tau > 0$ a monthly monetary equivalent of the time, learning and other participation costs common to both tiers. Utilities are

$$u_0 = 0, \quad u_F = vq_F - \tau, \quad u_P = vq_P - \tau - p_c. \quad (5)$$

These service levels are theoretical objects; they need not correspond to particular models, observed subscription labels or a measured technological frontier. The framework omits competing suppliers, employer-provided access, sharing and usage-dependent participation costs.

Write $\Delta q = q_P - q_F$. Direct comparison of all three options gives the thresholds for any use and paid use:

$$T_{D,c} = \min \left\{ \frac{\tau}{q_F}, \frac{\tau + p_c}{q_P} \right\}, \quad T_{P,c} = \max \left\{ \frac{p_c}{\Delta q}, \frac{\tau + p_c}{q_P} \right\}. \quad (6)$$

Thus a free option does not, by itself, make aggregate adoption independent of the paid price. A sufficiently attractive paid option can draw individuals directly from non-use.

Let κ_c be the population share with the connectivity and complementary resources needed to participate. Conditional on belonging to this group, assume

$$\ln v \sim N(\mu_c, s^2), \quad \mu_c = \mu_0 + \ln(y_c/y_0), \quad s > 0, \quad (7)$$

where y_c is a country income index and y_0 a reference value. This location-shift assumption is a stylisation, not an identity linking national GNI to household incomes or returns to AI. Defining the distribution conditional on eligibility makes the multiplication by κ_c explicit; applying the same distribution unconditionally would require an additional selection assumption. Holding qualities, participation costs and dispersion common, the population shares using any tier and the paid tier are

$$D_c = \kappa_c \Phi \left(\frac{\mu_c - \ln T_{D,c}}{s} \right), \quad P_c = \kappa_c \Phi \left(\frac{\mu_c - \ln T_{P,c}}{s} \right). \quad (8)$$

The assumption of a common τ is substantive. If participation costs vary as $\tau_c \propto y_c^\lambda$, the nested-regime adoption slope becomes $\varepsilon_\kappa + (1 - \lambda)h(z_D)/s$ even at a fixed free service level. Monetary time costs can rise with wages; the baseline assumption therefore rules out an economically relevant source of slope variation. Arbitrary heterogeneity in participation costs cannot generally be absorbed into v while retaining both threshold formulas and a common paid price.

Threshold ordering. If

$$\frac{p_c}{\Delta q} > \frac{\tau}{q_F}, \quad (9)$$

then $T_{D,c} = \tau/q_F$ and $T_{P,c} = p_c/\Delta q$. Within this regime, a marginal paid-price change leaves D_c unchanged, provided κ_c , qualities and participation costs do not change. If the inequality is reversed, both thresholds equal $(\tau + p_c)/q_P$ and paid price affects any use as well as paid use.

Proof. The free tier is preferable to non-use when $v \geq \tau/q_F$. The paid tier is preferable to non-use when $v \geq (\tau + p_c)/q_P$, and preferable to the free tier when $v \geq p_c/\Delta q$. These comparisons establish equation (6). Under condition (9), the three thresholds are ordered as $\tau/q_F < (\tau + p_c)/q_P < p_c/\Delta q$. Reversing the condition reverses this ordering, establishing both regimes. At equality the thresholds coincide. $\square$

Proposition 2 (Local income gradients). *Within the strict nested-threshold regime, define $z_D = (\mu_c - \ln(\tau/q_F))/s$, $z_P = (\mu_c - \ln(p_c/\Delta q))/s$, $h(z) = \phi(z)/\Phi(z)$, $\beta = \partial \ln p_c / \partial \ln y_c$, and $\varepsilon_\kappa = \partial \ln \kappa_c / \partial \ln y_c$. Then*

$$\varepsilon_D = \varepsilon_\kappa + \frac{h(z_D)}{s}, \quad \varepsilon_P = \varepsilon_\kappa + (1 - \beta) \frac{h(z_P)}{s}. \quad (10)$$

Uniform pricing gives $\varepsilon_P > \varepsilon_D$. Holding the local price level and hence the threshold values fixed, this ordering persists if $\beta < \beta^ \equiv 1 - h(z_D)/h(z_P)$, where $0 < \beta^* < 1$.*

Proof. Differentiating the logarithms of equation (8) yields equation (10), since $\partial \mu_c / \partial \ln y_c = 1$. The strict threshold ordering implies $z_P < z_D$. The lower-tail inverse Mills ratio h is strictly decreasing, so $h(z_P) > h(z_D)$. Subtracting the elasticities and solving for their equality gives β^* . $\square$

The comparison across β is local. One implementation compares schedules $p_\beta(y) = p_0(y/y_0)^\beta$ at their common anchor y_0 . Away from that anchor, changing the pricing schedule also moves the paid threshold; the proposition does not establish a global effect of a localisation policy. Nor does it establish welfare effects. Its substantive implication is conditional: with nested thresholds and the maintained distributional assumptions, a free tier can separate the determinants of initial adoption from those of upgrading.

In the reversed-threshold regime, the same calculation instead gives

$$\varepsilon_D = \varepsilon_P = \varepsilon_\kappa + \left(1 - \beta \frac{p_c}{\tau + p_c}\right) \frac{h(z)}{s}, \quad z = \frac{\mu_c - \ln((\tau + p_c)/q_P)}{s}. \quad (11)$$

There is then no interval in which the free tier is the individual's optimal choice. This limiting case makes clear why the empirical presence of free products alone cannot establish price independence. More generally, differences across countries in language support, time costs, service quality, selection into connectivity or the dispersion of valuations introduce additional terms into the gradients. These possibilities are substantively relevant and are not ruled out by an income regression. The framework should therefore organise questions about access while leaving these competing explanations open.

B Data construction and replication

Replication package. The package contains the raw extracts (World Bank API pulls, the Microsoft dataset at commit 507c316, Anthropic Economic Index files, Google ATLAS data, ITU price baskets, the App Store scrape, exchange rates), the processing scripts, the analysis, and the figure and table generators, together with a README that follows the template of the Social Science Data Editors. Every

number in the text can be reproduced from it: most are written to output/results.json by the analysis script, the sensitivity analyses are written by the scripts in revision_audit/ and publication_upgrade/, and the rest are simple calculations from those values or are read from the output tables, the processed data or the scripts. Re-running the pipeline from the shipped raw extracts reproduces all 865 entries of the results file and every table body byte for byte, and independent NumPy implementations of the key regressions match the reported estimates. The pinned Microsoft source, four Anthropic releases and the ITU workbook were also re-downloaded and re-processed, and the derived files were byte-identical to those shipped. What the package cannot do is replay the App Store extraction itself, because it stores the parsed listing records rather than the original pages; the plan-matching checks of Section 5.2 test the selection rules on those records.

App Store audit. Requests were made on 27 September 2026 (UTC) to <https://apps.apple.com/<cc>/app/id<app>> for ChatGPT (6448311069), Claude (6473753684) and Google Gemini (6477489729), using the two-letter codes of all 217 World Bank economies plus Taiwan. Of 654 requests, 500 returned local pages, 135 were redirected to the U.S. store and 19 returned “not found”. Prices were parsed from the displayed strings, including locale-specific separators and abbreviations (e.g. Indonesian “ribu” and “juta”). Four storefront-plan cells in which the observed page listed only an annual price were dropped. In three cells two different candidate prices are listed for the same plan: Google AI Pro in India and Türkiye, where one of five listings carries a lower price, and in Benin, where a 10 TB variant is listed alongside the standard plan. I use the modal price. Taking the lowest listed price instead changes the Google AI Pro elasticity by 0.0001.

Microsoft region-imputed economies. Six pools with identical four-period series: West Africa (Benin, Burkina Faso, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Sierra Leone, Togo); East Africa (Burundi, Eritrea, Ethiopia, Somalia, South Sudan, Sudan, Tanzania, Uganda); Central Africa (Cameroon, Central African Republic, Chad, Congo, Congo DR); Southern Africa (Angola, Madagascar, Malawi, Mozambique); the Guianas (French Guiana, Guyana, Suriname) and Venezuela; and Afghanistan, Tajikistan and Turkmenistan. These 36 economies coincide with those marked as region-imputed in Misra et al. (2025, Appendix Table 2). Because French Guiana and Taiwan are not World Bank economies, 145 Microsoft economies are matched and 142 have income data.

Anthropic index. Published AUIs are used for April and May 2026. For August and November 2025 and February 2026 the index is recomputed as usage share over working-age population share among geolocated countries with at least 200 sampled conversations, because the August 2025 published values include un-geolocated traffic in the usage denominator (scaling all values by a common factor of 0.843) and later releases publish counts rather than indices.

C Connectivity accounting

Define the ratio $R_c = D_c/I_c$, where D_c is Microsoft’s working-age AI-use share and I_c is the all-age internet-use share, so that $\ln D_c = \ln I_c + \ln R_c$ exactly. In the 109-country estimation sample no ratio exceeds one. The slopes on log income are 0.182 for internet use and 0.189 for the ratio, summing to the adoption slope of 0.370 up to rounding, so the connectivity component accounts for 49% of the

Table 13: Connectivity decomposition of the AI-use gap (country-specific estimates, Q2 2026)

Income group	N	Group means (%)			Log gap to high income			Connectivity share (%)
		AI users	Internet	AI / internet	Total	Connect.	Ratio	
High income	43	34.9	92.6	37.5	0.00	0.00	0.00	–
Upper-middle income	38	18.5	81.8	22.7	-0.63	-0.13	-0.50	21
Lower-middle income	25	12.2	59.6	23.6	-1.05	-0.51	-0.54	49
Low income	3	8.8	38.3	23.0	-1.34	-0.90	-0.44	67

Notes: $\ln D = \ln I + \ln(D/I)$ with D the AI user share (population 15–64) and I internet users (% of population, latest year). Log gaps are differences between group means of log values (log ratios of geometric means), so they differ from log ratios of the arithmetic means shown. The low-income row rests on three economies and is indicative only. The ratio D/I mixes age bases; no cap binds in the estimation sample.

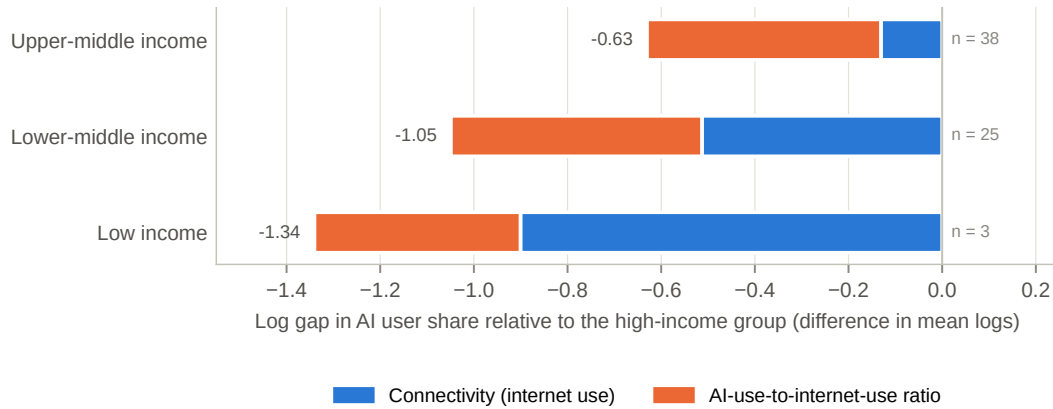


Figure 6: Decomposition of the log gap in AI use relative to the high-income group (difference in group means of log shares) into connectivity (internet use) and the AI-use-to-internet-use ratio. Country-specific Microsoft estimates, Q2 2026; the low-income bar rests on three economies.

gradient. This is an accounting decomposition: the denominator has a different population base and often a different year, internet penetration already enters Microsoft’s estimator, and holding income fixed the elasticity of AI use with respect to internet use is 0.05 (s.e. 0.20), far from the value of one that a pure connectivity channel would imply.

Across groups (Table 13, Figure 6) the connectivity share of the gap to high-income economies rises from 21% for upper-middle-income to 49% for lower-middle-income and 67% for the three low-income economies with country-specific data. The ratio of AI users to internet users is 22.7%, 23.6% and 23.0% from the upper-middle- to the low-income group, against 37.5% in high-income economies, and its income slope across non-high-income economies is -0.03 (s.e. 0.06). The comparison rests on group means, the low-income row contains only three economies, and the ratio mixes age bases; it is not a directly observed probability of adopting AI conditional on being online.

D Additional tables

Table 14: Income gradient of AI use with controls (log user share, country-specific, common sample)

	(1)	(2)	(3)	(4)	(5)
ln GNI per capita	0.3595 (0.024)	0.3554 (0.047)	0.3659 (0.063)	0.3566 (0.062)	0.3675 (0.067)
Internet users (%)		0.0004 (0.004)	-0.0006 (0.004)	0.0002 (0.004)	0.0016 (0.005)
Tertiary enrolment (% gross)			-0.0015 (0.002)	-0.0011 (0.002)	-0.0006 (0.002)
Urban population (%)			0.0064 (0.003)	0.0063 (0.003)	0.0040 (0.004)
Population 65+ (%)			-0.0077 (0.008)	-0.0083 (0.008)	-0.0011 (0.011)
Paid online merchant digitally (%)				-0.0022 (0.002)	-0.0024 (0.002)
Region fixed effects	No	No	No	No	Yes
Observations	97	97	97	97	97
R^2	0.69	0.69	0.72	0.73	0.74

Notes: OLS with HC3 standard errors. Controls measured in percent; Findex 2025 for digital payments.

Table 15: Income elasticity of the Anthropic AI Usage Index over time

Window	All qualifying economies				Common sample ($N = 111$)	
	Elasticity	(s.e.)	N	R^2	Elasticity	(s.e.)
Aug 2025	0.676	(0.041)	114	0.71	0.676	(0.043)
Nov 2025	0.690	(0.036)	115	0.76	0.685	(0.040)
Feb 2026	0.750	(0.035)	115	0.80	0.741	(0.037)
Apr 2026	0.767	(0.033)	113	0.84	0.769	(0.033)
May 2026	0.741	(0.028)	120	0.85	0.734	(0.031)

Notes: OLS of log AUI on log GNI per capita (2025); economies with at least 200 sampled conversations (Aug 2025–Feb 2026) or with published values (Apr–May 2026). Common sample: economies qualifying in all five windows; change Aug 2025 to Feb 2026: 0.064 (clustered s.e. 0.018); Feb to May 2026: -0.006 (0.016).

E A \$500 reference-fee scenario

The tariff change of 29 September 2026 (Section 2.1) supplies an observed U.S. web fee for a new high-usage plan but no local prices, taxes or eligibility conditions for it. Table 16 therefore applies a common monthly fee of \$500 to the national-income sample as a scenario, alongside the \$20 and \$200 reference fees. Each country’s ratio is computed before taking group medians.

Table 16: Annual reference fees relative to GNI per capita

Income group	N	\$20 per month	\$200 per month	\$500 per month
High income	75	0.65	6.48	16.21
Upper-middle income	58	3.09	30.87	77.17
Lower-middle income	47	8.73	87.27	218.18
Low income	21	28.92	289.16	722.89

Notes: Medians of 1200p/y in percent across the economies of each group. The \$500 column applies the web fee of the Pro 500 tier announced on 29 September 2026 as a common fee; no local prices for that tier were observed.

At \$500 a month the annual fee is \$6,000, and the 2% benchmark corresponds to a GNI per capita of \$300,000, which no economy reaches. Because the burden is proportional to the fee, the \$500 burden is exactly 25 times the \$20 burden in every country; the scenario scales the levels without changing the relative cross-country ratios or adding any variation that could identify a price effect. It also says nothing about whether free or lower-cost tiers suffice for a given task.

F A protocol for recording access conditions

A recurring fee does not determine the cost of completing useful work. A reproducible observation of access conditions should record the country, observation time, billing channel, provider, plan and billing period; the original currency and tax convention; listing and checkout status; subscriber cohort; included allowance and reset rules; model and tool eligibility; and overage terms. Missing terms should remain missing rather than be assigned a provider-wide constant. The replication package supplies a field dictionary for these observations (publication_upgrade/access_measurement_fields.csv). It is a proposed extension of the present audit, not a newly collected dataset.

For a prespecified workload B , quality rule q and deadline d , define the feasible set $\mathcal{F}(B, q, d; c, t)$ of service configurations that actually meet all three requirements in country c at time t . The minimum cash expenditure is

$$C(B, q, d; c, t) = \inf_{a \in \mathcal{F}(B, q, d; c, t)} \text{cashcost}(a).$$

The comparison must state whether free, API, institutional and local-model configurations are included and how fixed hardware or subscription costs are allocated. Feasibility requires task-level observations of correctness, language, model version, wall-clock completion time, retries, billed use and failures. This paper does not observe those quantities and does not estimate C . A prospective evaluation should report total expenditure and success rates alongside cost per successful task, since excluding failed attempts would understate costs. The protocol distinguishes measured subscription commitment from a future quality- and deadline-constrained access outcome.